AETHER System: User Input-to-Output Flow

The diagram photo represents the user input-to-output processing pipeline within the AETHER system using its modular structure.

1. UserInput: The user types a feeling or phrase, such as “I feel tired.”

2. Infernia: InputHandler (parse) — The input is parsed into an Intent object containing the type, content, and confidence level.

3. Intent Creation — An Intent object is created and passed into the system for processing.

4. Infernia: Controller — The controller routes the intent by applying rules (e.g., logging or timing constraints).

5. Aletheia: CLU (process, decideAction) — The Cognitive Logic Unit (CLU) decides how to proceed—whether to request context, trigger memory recall, or evaluate emotional state.

6. Mnēsis: MemoryEngine (shortTerm, longTerm) — If context is needed, the CLU queries the MemoryEngine, which retrieves relevant data (e.g., past interactions or stored emotions).

7. Context Returned — Contextual data is returned to the logic system for informed reasoning.

8. Reverie: EmotionSystem (moodState) — The system evaluates the user's emotional state, checking if the mood is tagged as LOW_ENERGY.

9. Conditional Evaluation (Diamond Node) — A decision is made:
   - Yes: Mood is low energy → Display a suggestion like “Suggest rest/music.”
   - No: Continue to generate a more complex response.

10. Reverie: InterfaceRender (text/audio) — The chosen output (text or audio) is rendered for the user.

11. UserInterface — The system communicates the result: e.g., “Suggest rest/music.”

12. Response Object — A Response object is created that includes:

- Text
- Priority
- Emotion tag

13. MoodGauge: Tired — The user's mood state is updated (e.g., logged as "Tired").

